IXXN341,
Design For Health.



IR Remote plugin device for MND Patients and Low Mobility users.

Campbell Smith

Report: Introduction

Motor Neuron Disease, or MND, is a neurological disease that stops the communication betweenn the brain and the motor neurons, the nerves that control body movement.
The Disease is proressive, and terminal, resulting in patients typically loosing body movement, swallowing, talking, and breathing (Basic Facts About MND - MND NZ, n.d.).



[Fig 1. Photograph, SmartBox Lumin-i]

A common technology used by those with the disease in New Zealand are Augmentative and Alternative Communication devices, AAC, with eye tracking camera attachments, Primarily from SmartBox. These devices work as a typical windows device, and give people access to communication software. In some of the newer devices they include IR controller properties which can be programmed to work with common household devices such as TVs, but are often neglected due to their niche nature, and are put into the background of the devices software, making them difficult to set up for the average user.

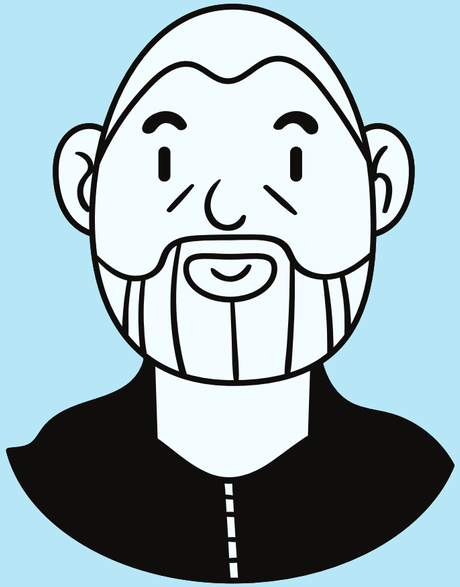
Report: Introduction

My idea is to create a modular plug in for older devices to give them IR capabilities, and develop a simple software app for users of these devices so the barrier to entry in setting them up is lowered.

My key thinking for this was my dad's struggles with the IR technology in his device, which he wanted to use to reduce caregiving workloads for tasks such as controlling his rooms heatpump. Once we found out his device had IR capabilities, and it's attached software had unctions built for it, I spent multiple hours doing research, and playing with his tablets software due to the complexity the set up required.

Once set up, the organisation who had supplied his technologies, TalkLink, on a returning check in visit found out we had been able to set up the IR controls and we had to walk them through step by step on how it worked as it had been something they were unable to do, but had been asked about by others with motor nueron disease.

Personae #1, John Doe



- Age, 65,
- Male,
- MND Paitent,
- No Family History,
- Prior Occupation, Sales
- Family Connections,
  2 Children in 30s, Wife Jane Doe

Skills and Abilities

Strength

Dexterity

Constituion

Intelligence

Wisdom

Charisma

Technology

Characteristics,

- Lacks physical capabilities with limited movement,
- Witty, and Clever,
- Organised,
- Motivated by interests,
- Not technologically savy, but is capable in basic technology use due to work,

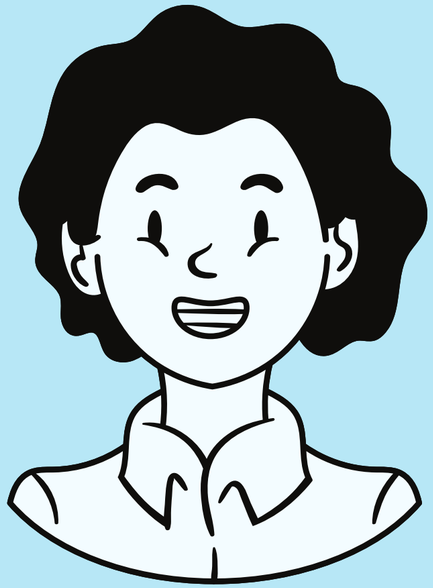
Goals,

- To be able to control basic household devices by self using his existing assistive technologies,
- Reduce workload on partner through indipendance,

Challenges,

- Not ICT Savy, so unable to devise a solution by self,
- Reliant on one Windows based device to use technology,
- Reliant on another person for any physical set up,
  - Experiencing mild cognitive decline,
    - Eye fatigue effects the longevity of technology use.

Personae #2, Jane Doe



- Age 62,
- Female,
- Occupation, Sales
- Family Connections,
  2 Children in 30s,
  Husband

Skills and Abilities

Strength

Dexterity

Constituion

Intelligence

Wisdom

Charisma

Technology

- Characteristics,
  - Patient,
  - Not the most ICT savy but is ready to ask children,
  - Dedicated to caring for husband,
- Goals,
  - Give as much autonomy to John as possible,
  - Reduce own workload for menial tasks to focus on bigger issues,
- Challenges,
  - Low Technology Confidence,
  - Reliant on others for technology advice,
  - Very busy time wise, and mentally taking care of John,

Personae #3, Dr. Smith



- Age 45,
- Female,
- Occupation, Occupational Therapist,
- Some experience with MND Patients,

Skills and Abilities

Strength

Dexterity

Constituion

Intelligence

Wisdom

Charisma

Technology

Characteristics,

- Experience working with assistive technology just above surface level,
- Paitent,
- Well informed about paitents and caregivers daily lives,
- Decently ICT Savy but also has connections to IT medical departments,

Goals,

- Give Paitents a sense of independance,
- Ensure managible workload to caregivers,
- Ensure both paitents, and caregivers, are happy with care,

Challenges,

- Limited time with each paitent,
- Frustrated with lack of easily accessed solutions,
- Limited knowledge on solutions to only what is current and funded,

May be vaguely aware of other solutions,

Interview Exerpts

What technologies to assist you do you currently have access to?

John Doe:

I do have the eye-gaze device – got that set up through the MND service not long after diagnosis..... It's mainly for communication, talking to people, that sort of thing, using the AAC software on it..... So I've got a good "voice," in a manner of speaking, but no real way to act on the house around me.

Dr. Smith:

.....most MND patients are set up with an eye-gaze device and AAC software fairly early in the disease trajectory.....What's less consistent is whether that same device gets extended into environmental control – some AAC software has that capability built in or as an add-on, but it depends on the platform, and a lot of patients and even some clinicians aren't aware the functionality exists.

How well can you navigate new software, would you be able to direct someone else confidently in using the same software?

John Doe:

Navigating new software myself – slower than it used to be, but I can manage if it's laid out sensibly.....I can figure most things out....., but I need to do it in short stints rather than one long sitting. Give me a day of tinkering and I'd have a decent handle on it.

Dr. Smith:

For myself, reasonably well.....I can usually work out new software within a session or two.....A day of tinkering would generally be enough for me to give a solid walkthrough to a colleague or caregiver.....The bigger issue isn't really my own learning curve, it's that I've got limited time per patient, so even if I could confidently teach the software, fitting that teaching into an already packed session is its own challenge.

Key Takeaways

John has Access to AAC software and eye tracking which he uses only for communication, and isn't aware of how it could be used outside of that function.

Software needs to be incredibly simple and intuative on the user facing end, so is easy to set-up and explain either by self or in a short period of time with a professional.

Interview Exerpts

How hard is setting up any new devices for you?

John Doe:

Physically, I can't set one up myself ..... Now the actual doing has to be someone else's hands.....What I can still do is think through it, tell someone what I want set up and roughly how.....It just takes patience on both ends, and it's slow going explaining every little step rather than just grabbing the thing and doing it myself like I used to.

Jane Doe:

Setting up a.....device is one of those things I dread a bit.....I can follow instructions if they're written out simply, step by step, but if something goes wrong or it asks me something unexpected I get a bit lost...

How in depth do you like understanding your devices?

John Doe:

Before all this, I liked knowing how things worked properly....., Now though, with the eye fatigue and everything taking longer, I've had to let go of wanting to know every custom option and plugin. I want to understand enough that I'm not just a passenger in my own house — I'd like to know roughly what's possible so I can ask for it

Dr. Smith:

Professionally, I sit somewhere just above surface level..... I understand what a device or piece of software is capable of, and I can usually configure the standard accessibility settings.....If something needs proper deep customisation, that's when I loop in our IT medical contacts. What I really want, for both myself and for families like Jane's, is documentation and setup that sits at a "confident intermediate" level — enough transparency that I understand what's happening under the hood well enough to troubleshoot common issues and explain it to a caregiver, without needing to be a developer to get there.

Key Takeaways

Clear set-up instructions.

Easily understandable to users, with some background on additional background so they are aware of other use cases.

Easily understandable for professionals for set-up and troubleshooting purposes, with guidance for other use cases and their set-up.

Interview Exerpts

What would be you main wants out of this product?

John Doe:

Independence, mostly – being able to turn on the telly.....without having to get someone's attention and wait for them to be free. Even something small like that matters a lot when you can't do much else for yourself. I'd want it to work through the eye-gaze since that's already my main way of doing anything, and I don't want to have to relearn a whole new system or hold yet another piece of hardware.

Jane Doe:

.....simple enough that if something does go wrong I've got a fighting chance of sorting it myself, or at least explaining the problem clearly to whoever I ring for help.

Dr. Smith:

From my side, I want something that genuinely increases patient autonomy without creating extra burden for caregivers or for me.....I'd also want it to be something I can actually get funded and supported through existing channels, or at least something simple enough that funding isn't a barrier....

Does the solution need to look good? If so something like a stand out art piece, or more sleek that blends in?

John Doe:

Sleek and blending in, for me.....I've had enough of my house filling up with "medical looking" equipment..... Something that just looks like a normal bit of tech sitting on the side,

Dr. Smith:

From a design standpoint, I'd lean towards sleek and unobtrusive over a standout piece, and this comes up a lot with home-based assistive tech generally – patients and families often prefer things that don't visually signal "this is a disability aid" sitting in the lounge.

Key Takeaways

The solution needs to provide independence and something frictionless that requires less time and energy than is currently needed by human intervention.

The product needs to blend in, it shouldn't be bold or medical looking, more like something you'd expect to see plugged into a device.

Affinity Diagrams

What features should existing AAC Devices have?

- Tactile Features for Early Stage MND Patients to retain Mobility and a sense of normality.
- Modular design to fit and adapt to individual needs.

Physical Changes

- Simple interfaces for not ICT Savy users.
- All digital set up should be able to be set up by user without external intervention.
- Clear Instructions and directories for information on set-up and troubleshooting.

Interface Changes

What Should be prioritised in a AAC device plugin?

- Non-Intrusive design → Clear as to purpose

Physical Properties

- Clear instructions
- Simple to set up, low barrier to use

Simplicity

- Simple user interface → Minimal physical intervention required

Patient Autonomy Centered

What is the key issues around existing enviromental controls for MND patients in Devices

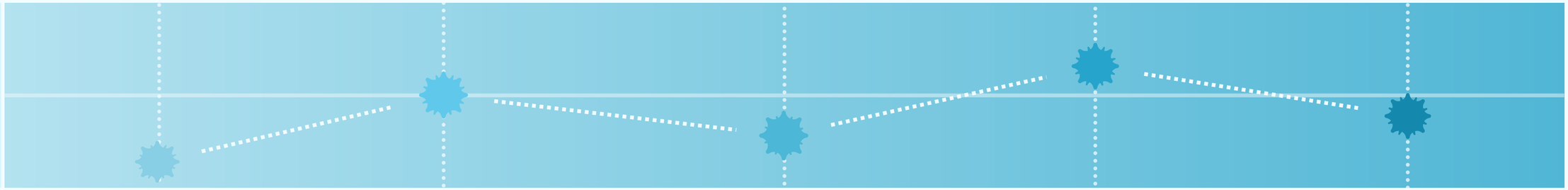
- Difficult to Access
- Hard to set-up/program
- Nothing user friendly exists
- Hidden in existing devices

Clarity

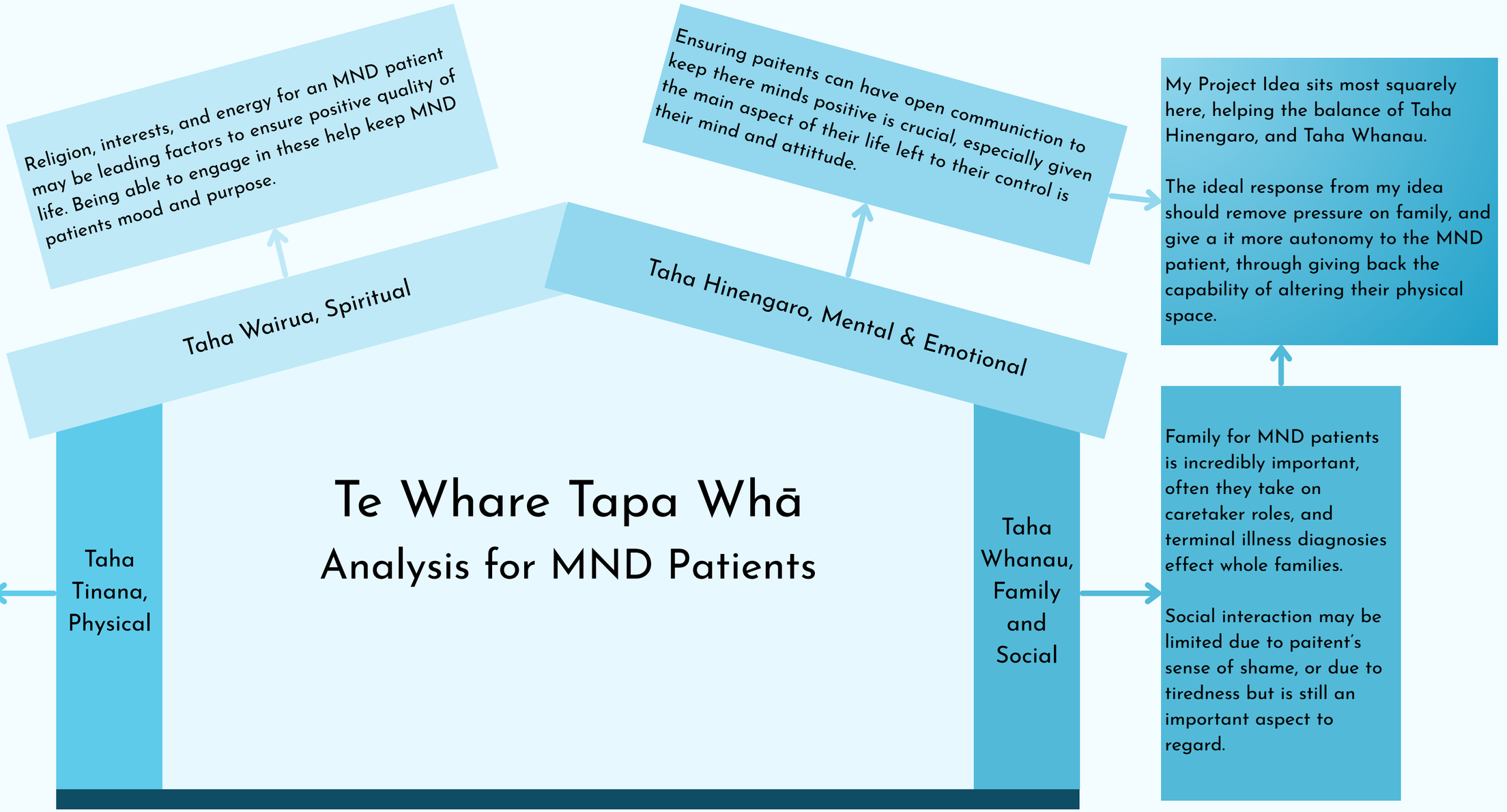
- No specialised IT personal in this area
- Patients with limited energy to exert on set up
- Hardware is confusing to non tech people
- People aren't aware the hardware exists and what it can do

Lack of Expertise

Customer Journey Map: MND Patient Diagnosis - Technological Intervention

	Phase #1	Phase #2		Phase #3	
	Paitent Diagnosis	***Intial Support Communication***	***Medical Device Supply and Set-up***	***Device Use.***	***Troubleshooting***
Patient Goals	Understanding what is going on, and what future may look like.	Setting up a medical support foundation and understanding solutions out there.	Basic understanding of supplied devices, how they work, what they can do.	Becoming effective at using adaptive technologies to perform tasks they were able to do prior.	Finding solutions to problems with the technology as they come up, and having open access to medical IT support.
Patient Actions	Neurological testing, multiple doctors appointments.	Meeting lots of individuals, socially with MND experiences, and medically with full support team.	Devices are delivered to home along with professionals from medical device supplier to show how to use everything.	Patient may spend a week or two with new Devices learning how they work for them and finding any pain points or pros.	Patient has to trouble shoot any technology issues with medical team and navigate around compromises to their accessibility.
Patient Thoughts	Shock, Fear, Confusion, Sadness.	Lots of people and things going on at quite a quick pace.	Can be intrusive with the ammount of people coming into the home, and also overwhelming with the ammount of new things to learn.	Exciting as issues are addressed with technology, but can also be frustrating if there are IT issues as these devices become a life line.	Frustrating that time may be wasted for small issues.
Patient Experience					
Patient Pain Points	Upsetting time for patient and family, unfamiliarity for what the process will look like.	Can be overwhelming if not properly supported.	Time consuming, lots of information on new things to take in and remember.	Lots of information and new things to learn, with the patient not having a lot of time to invest.	Who to contact for what, people not knowing everything about device issues and having to contact others.
Patient Improvements	Ensuring patient comfort, and warmth from all medical team involved.	One direct commuunication point, and some form of organisational directory on who to contact for what.	As quick and efficent as possible.	Streamlined simple devices, with intuative interfaces that don't require too much new learning.	Links back to confusing new devices, and unintuative interfaces.

Māori Health Model Reflection: Te Whare Tapa Whā



Empathy Map: John, Looking at an ACC Device with IR Controls

Says

I want to be independant

I want to control my encviroment

I want it to work first time

I want to put less stress on my family

I don't want it to get in the way

Thinks

One less thing I need help with

Will this work when I'm tired

I hope this doesn't slow me down elsewhere

I don't want turninng the TV on to feel like a chore



Does

Using it the same way he talks

Test the device when others are around to make sure he can do it on his own

Keeps on screen alongside talking so it is easy to navigate when tired

Pauses use when tired

Feels

Relieved, one less thing to put on Jane

Anxious, it may not work all the time so he wants a back up

More like himself

Proud, that he can work his enviroment himself once again

Service Blueprints

MND AAC Patient Device use

Stakeholders

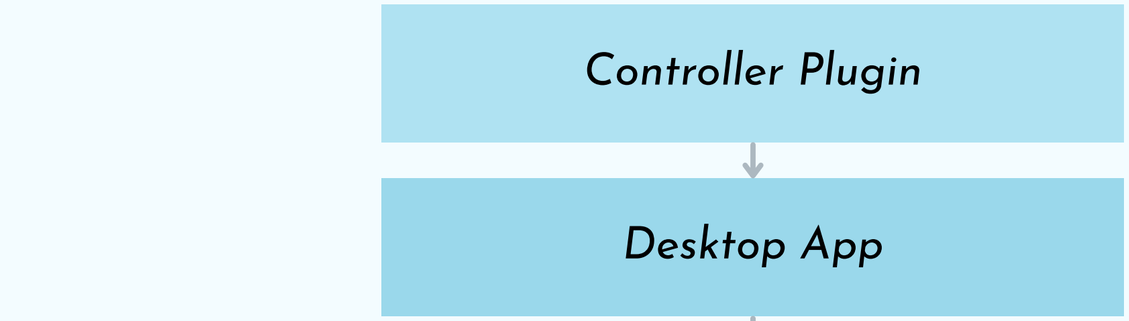
- MND Patients
  - Key Stakeholder, main user of device, limited mobility, any physical activity requires support, or another person to complete action
- Caregiver,
  - Typically a family member or Partner, often taking on all day to day in home tasks and care of patient
- Medical Team,
  - Includes, Neurologist, GP, Gastroenterologist, Community Nurse, Physiotherapist, Ocupational Therapist, External Care Giver, MND NZ Representative.
- TalkLink,
  - Technology company that provides AAC devices and software to MND Paitents and others publicly.

Time	Diagnosis - Month 0	Initial Set-up - Month 1-3	Living with Diagnosis - Months 3-?	Late Stage	Device Return
Location	Hospital	In-Home		Hospice Care	
Physical Evidence		AAC Device arrives in home with stands and attachments	AAC Device is used in home, with attachment accessories figured out	Device is adapted to new setting for user to still use	Device is stripped for new user
Patient Actions	Processing Terminal Diagnosis; Doing own research; Putting things in order, talking to family & friends	Discussions with Medical Team about needs and wants; Learning through trial and error about what device can do for them and how to use it	Discussing discoveries, and more wants/needs for technology with medical team	Troubleshooting device errors as they arise and using technology daily	
Caregiver /Family Actions		Discussing with Paitent and medical team; Learning new technology with patient to be able to assist where needed	Troubleshooting device errors, Performing Care tasks, intervening to help with device as needed.		Clears device of personal information
Medical Team Actions	Nuerologist contacting MND NZ, and other Medical staff to fill out team for patient	Medical Team doing intial reach out and support, gathering what the patient and family need.	Frequent contact with paitent and family on a weekly/daily basis	Hand over to Hospice Care, still working with patient and now Hospice to ensure quality care.	Familty Support
Talk Link Actions		Medical Team talking to TalkLink around communication devices, and patient needs.	Specific device help, and device exploration with patients to develop a better understanding of needs and wants to cater better to ffuture users.		Device Collection
SmartBox		TalkLink Purchases AAC Device and software from SmartBox	SmartBox offers training and guides to their devices for TalkLink and Paitent.		

Initial Sketches

Note: Not final colour sheme, the colours here were just choosen to match report

Tool Structure



User Dashboard

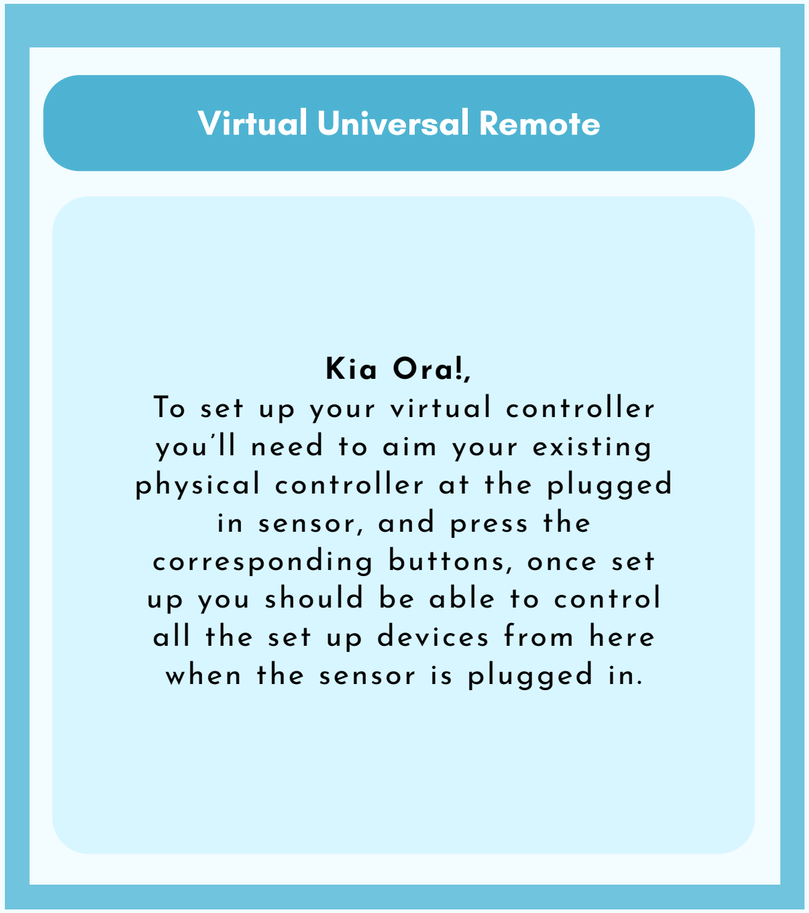
- Simple Dashboard,
- Quick Set up,
- Drag and Drop button adjustment,,
- Designed with MND patient use in mind
  - Big buttons, small screen so they can open alongside other software

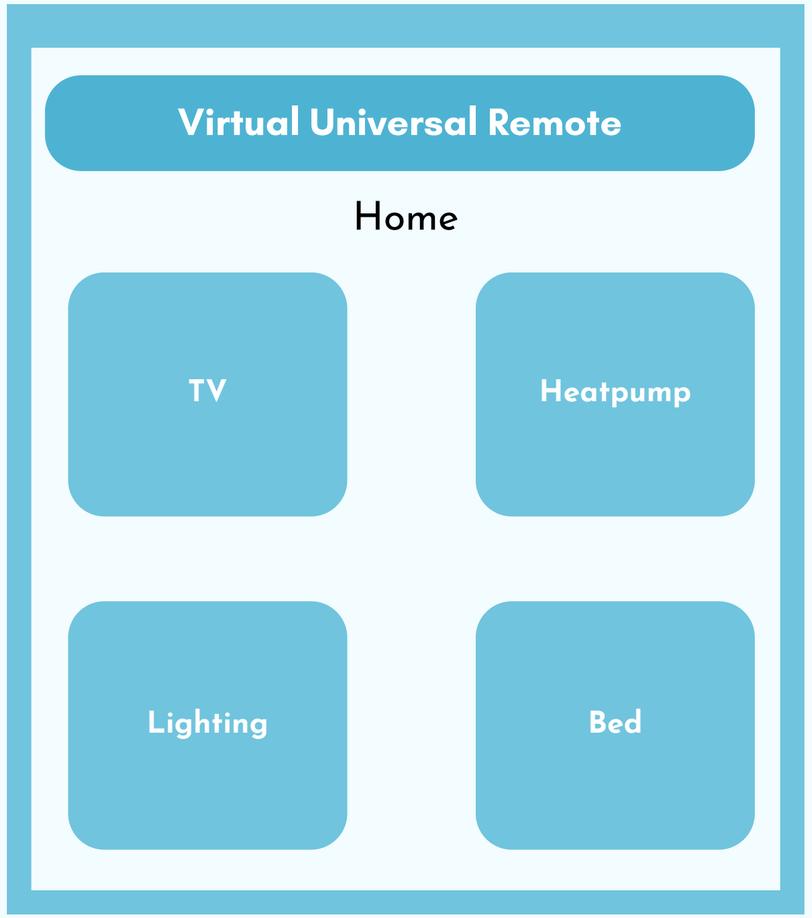
Developer Mode (?)

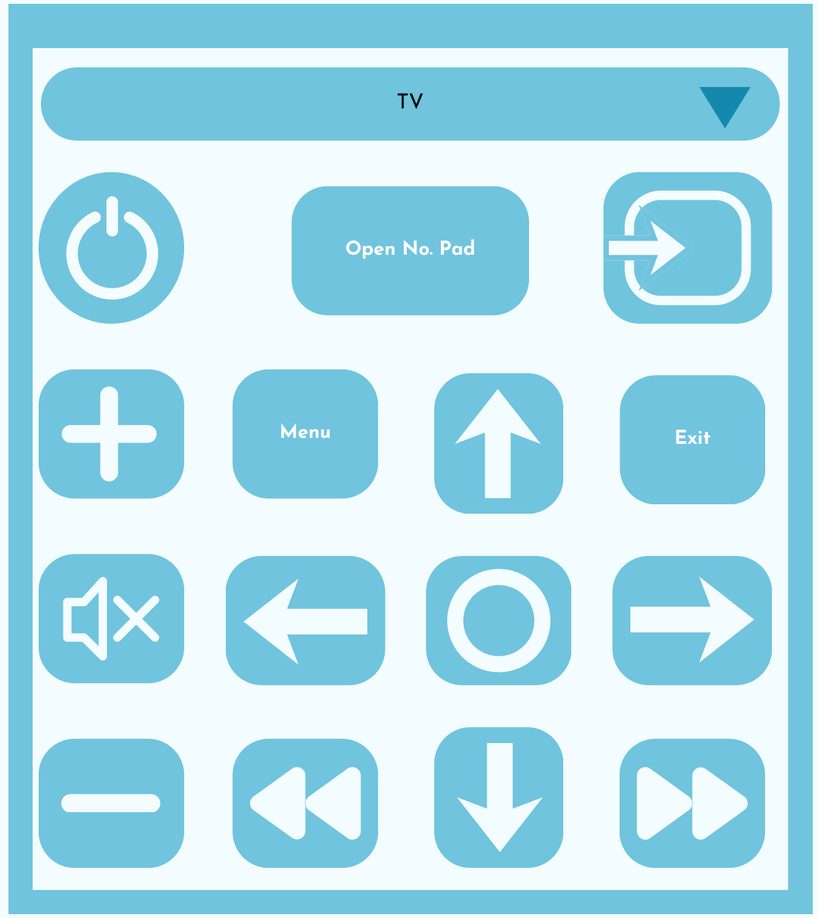
- More expansive dashboard,
- Designed for more complicated set up for more ICT savy users,
- Less navigatible by MND Patients due to smaller buttons

Initial Sketches: User Dashboard

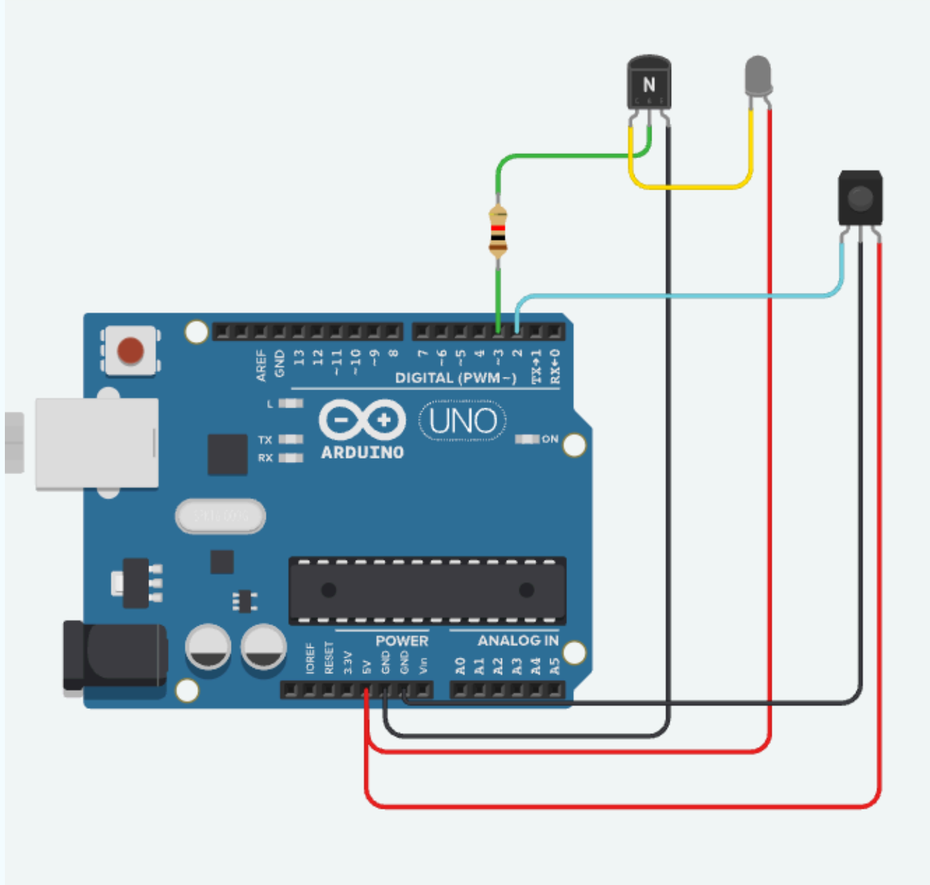
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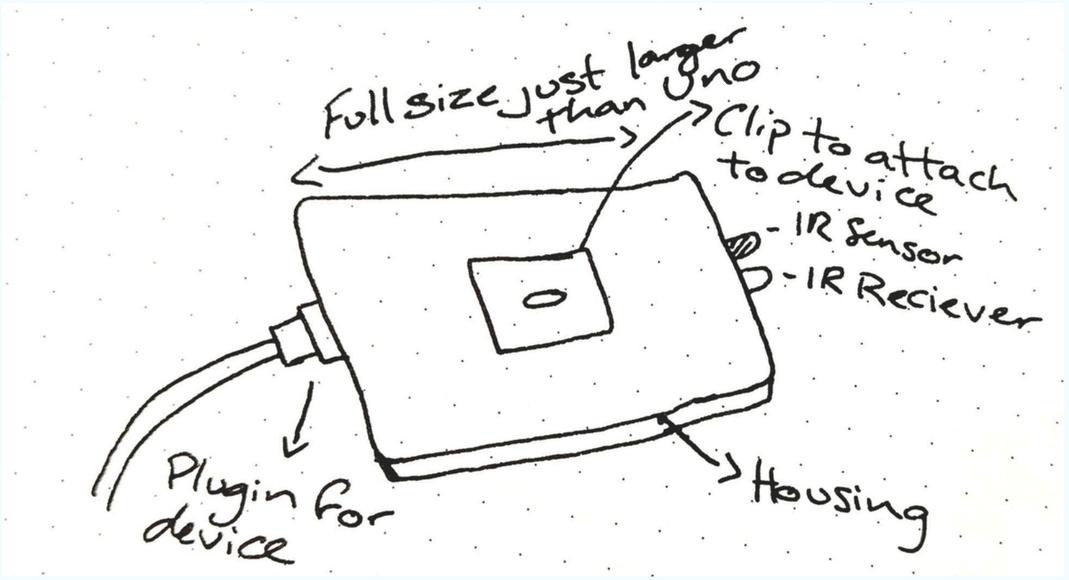


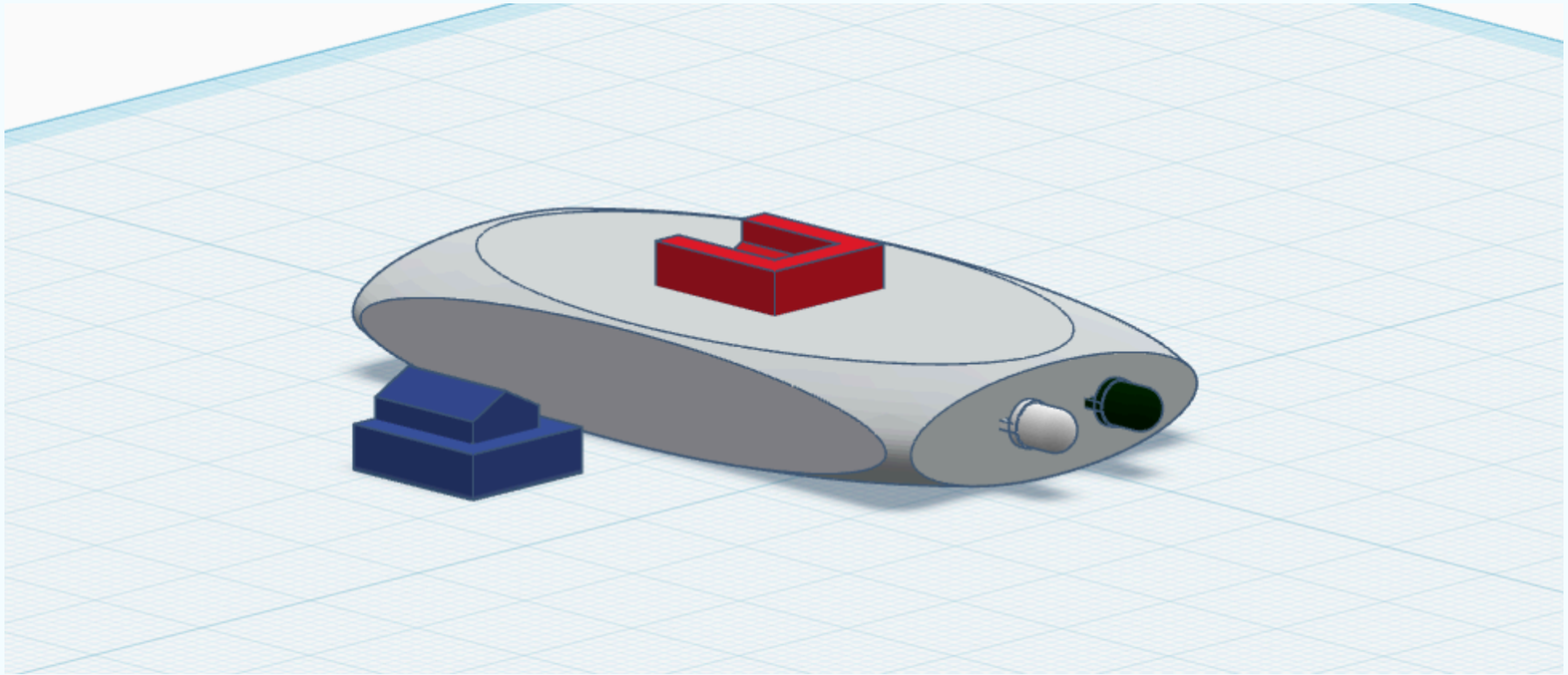




Initial Sketches: Physical Plugin







Device needs, Microboard, Transitor, IR Sensor, IR Reciever, and Resistors.

Report: Conclusion

I have come into this project with a background familiarity of MND due to my late father's diagnosis, looking through the methods I have come out with a handful of measures to apply to my design, those being,

- Simple Interface,
- Large buttons,
- Sleek hidden physical design,
- Hidden Developer options,

The key goal of this product is to allow and improve MND Paitents autonomy, through allowing personal enviromental control, aka letting them choose the Aircon temperature without having to call a caregiver or family member to do it for them.

This technology does already exist in some AAC devices, but is underutilised due to it's complicated set up nature, and developers hiding it behind settings, through the additional desktop app I plan on making idealy it would be able to sync up with these existing sensors and bring them to the forefront, allowing those with existing devices to use them to their full potential.

Bibliography

Cited References

Basic facts about MND - MND NZ. (n.d.). Motor Neurone Disease New Zealand. https://mnd.org.nz/about-mnd/what-is-mnd/basic-facts-about-mnd/

Background for Personaes based on stories shared with MND NZ, and personal experience

MND NZ. (n.d.). *Common difficulties*. Motor Neurone Disease New Zealand. https://mnd.org.nz/support-information/living-with-mnd/common-difficulties/

MND NZ. (n.d.). *Your stories*. Motor Neurone Disease New Zealand. https://mnd.org.nz/support-information/living-with-mnd/your-stories/

AI Disclosure

AI was used for the interview portion, and further brainstorming for the Empathy map based on the Personae shared. All writing and graphics, unless stated, are my own.

In-Built Canva Graphics Used



Images Used

Fig 1. [Photograph, SmartBox] https://thinksmartbox.com/try-our-products/

The Colours used in this Report are based on the brand Identity of MND NZ, which comes from the cornflower, the symbol for MND